· U C ·

Complex Systems

2023/2024

Normal Exam 24 of July 2024 Time: 1h30

Name: **Number:**

This is a closed books exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Question 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. If applicable, you may use a concrete example to **help you** to explain the concept.

a) Small World.

Your answer:

b) Radius (in a graph).

Your answer:

c) In order to be in a chaotic regime, a dynamical system must possess four characteristics. Identify them and explain their meaning.

Your answer:

Question 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

a) In agent-based modelling, the Moore's neighbourhood is preferable to the Von Neumann neighbourhood.

Your answer:

b) In terms of robustness it is better to have a system modelled as a small world network, rather than a system whose model is a scale free network.

Your answer:

Question 3 - Solve 25 points

Consider a 1D cellular automata with neighborhood of radius 1. Admit that someone said that he/she observed the following state transition:

$$t : 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 0$$

$$t + 1 : 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0$$

Is this situation possible? Justify your answer. What was the transition rule used and, if erroneous, correct it and present the new result.

Your answer:

Your answer:

Question 4 - Explain 25 points

In class we presented and studied a model for a predator-prey ecosystem based on differential equations. Thinking about this model and its components, in what way an agent-based approach or a network-based approach is more appropriate? Justify. Do not forget to describe these two models.

Your answer:

Your answer:

Question 5 - Model 20 points

During the course we discussed the use of the SIR (Susceptible-Infected-Recovered) model to analyse the spread of infectious diseases. This model was first created to use differential equations. However, the model can also use other approached like an agent-based one.

a) Describe an agent-based version of the SIR model. What advantages do you find in using this approach?

Your answer:

Your answer:

b) Now consider that agents can travel by plane. Describe how you can augment the model to include this.

Your answer: